



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MECHATRONIC SYSTEMS
NQF LEVEL 2

28 November 2025

This marking guideline consists of 6 pages.

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QUESTION 1

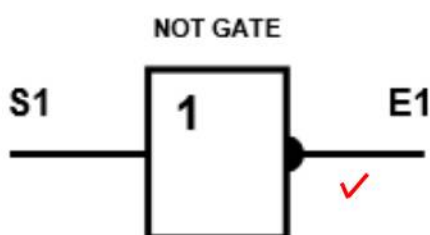
- 1.1.
- Chemical energy✓
 - Solar energy✓
 - Wind energy✓
 - Wave energy✓
 - Mechanical energy✓
 - Magnetic energy
 - Light energy
 - Nuclear energy
 - Thermal energy
- (Any 5 × 1) (5)
- 1.2
- 1.2.1 Basic system✓ (1 × 1) (1)
- 1.2.2 Sensors ✓ (1 × 1) (1)
- 1.2.3 Processor (control circuit) ✓ (1 × 1) (1)
- 1.2.4 Actuators ✓ (1 × 1) (1)
- 1.3
- The diagram shows a rainbow on the left and a horizontal bar on the right. The rainbow is labeled 'HOT' with a red checkmark. The horizontal bar is labeled 'COLD' with a red checkmark.
- (2 × 2) (4)
- 1.4 Carbon brushes ensure reliable operation of the DC motor✓ by maintaining continuous electrical contact.✓ Brushes help optimise motor performance by minimising electrical losses and reducing wear on moving parts.✓ Carbon brushes contribute to extended motor life by reducing wear and tear on the commutator and other components.✓ (4 × 1) (4)
- 1.5 The worm and wheel gear transmission is used to transmit motion and power between two shafts that are at right angles to each other.✓ The worm gear transmits power to the wheel gear, which multiplies torque,✓ enabling the system to handle heavy loads.✓ (3 × 1) (3)

[20]

QUESTION 2

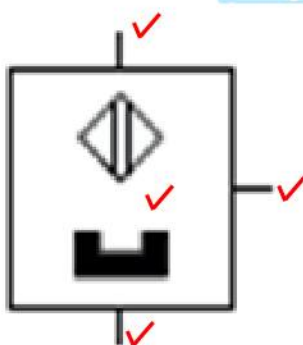
- 2.1 2.1.1 Device that compares a measured value with that of a preset value.✓✓ (2 × 1) (2)
- 2.1.2 Table consisting of all possible input combinations and resulting output.✓✓ (2 × 1) (2)
- 2.1.3 A switch is a device used to open and close an electric circuit.✓✓ (2 × 1) (2)

2.2



(3 × 1) (3)

2.3



(4 × 1) (4)

- 2.4
- Identify the problem or opportunity✓
 - Gather and analyse relevant facts✓
 - Determine suitable hypothesis or alternative✓
 - Evaluate and select the most appropriate hypothesis or alternative✓
 - Follow up on the decisions taken✓
- (5 × 1) (5)

- 2.5 2.5.1 Computer numerical control✓ (1 × 1) (1)

- 2.5.2 Computer integrated manufacturing✓ (1 × 1) (1)

[20]

QUESTION 3

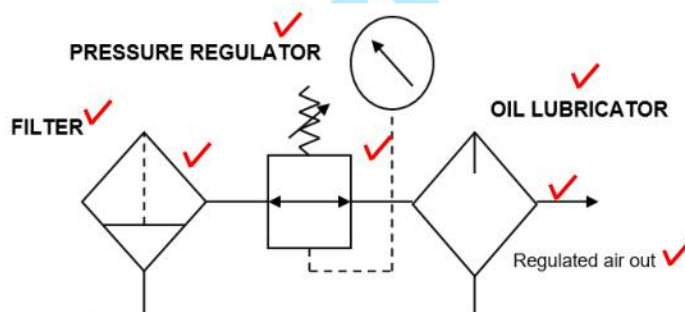
- 3.1 Compressed air is ordinary air that has been forced into a small space under pressure.✓✓ (2 × 1) (2)

OR

Compressed air is atmospheric air that has been pressurised to a higher pressure than normal atmospheric levels, usually using a compressor.✓✓

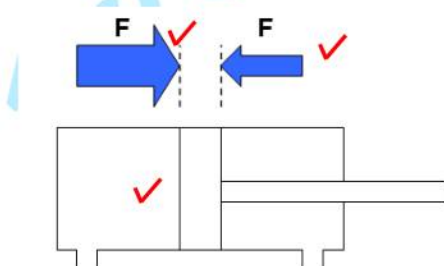
- 3.2
- Air is available in unlimited quantities practically everywhere✓
 - Air can be transported in pipes even over long distances✓
 - Compressed air can be stored in a reservoir and can be transported ✓
 - Compressed air is relatively insensitive to temperature fluctuations, thus ensuring reliable operation, even under extreme conditions✓
 - Compressed air offers no risk of explosion or fire
 - Un-lubricated exhaust air is clean, therefore air that escapes through leaking pipes or components does not cause any contamination
 - The operation of components is simple in design and construction and therefore relatively inexpensive
 - Compressed air is a very fast working medium, thus enabling high speeds to be attained
 - Pneumatic tools and operating components can be loaded to the point of stopping and is, therefore, overload safe (Any 4 × 1) (4)

3.3



(7 × 1) (7)

3.4



(3 × 1) (3)

- 3.5 A two-pressure valve is used to ensure that an output is activated only✓ when two input signals are present at the same time✓ as it performs a logical AND function in pneumatic systems.✓ To ensure that both hands are used to operate a machine, preventing injury.✓ (4 × 1) (4)

[20]

QUESTION 4

- 4.1
- Fuses✓
 - Circuit breakers✓
 - Overload relay✓
 - Current limiter✓
- (4 × 1) (4)
- 4.2
- 4.2.1 To ensure that a machine is restarted manually✓ if the voltage is lost due to a power failure.✓
- (2 × 1) (2)
- 4.2.2 E/stops are used so to prevent injury or damaged to machinery✓ when a fault or hazardous condition arises.✓
- (2 × 1) (2)
- 4.2.3 K1 in line 5✓ and K2 in line 7✓
- (2 × 1) (2)
- 4.2.4
- The circuit is a basic interlocking man/auto selector control circuit.✓
 - ON/OFF PB switches a system on/off✓
 - E/STOP PB activated when emergency exists switches system power OFF✓
 - START PB starts the system✓
 - RESET PB resets the system✓
 - MANUAL SELECTOR PB selects manual operation✓
 - AUTO SELECTOR PB selects automatic operation✓
- (7 × 1) (7)
- 4.3 24 to 48 volts.✓
By means of a step down isolating transformer ✓ and rectifying diodes.✓
- (3 × 1) (3)
- [17]**

QUESTION 5

- 5.1
- The filter is dirty✓
 - The pressure reducing valve is incorrectly set✓
 - The mist lubricator level control is not set properly✓
 - The shut-off valve is still closed✓
- (4 × 1) (4)
- 5.2
- Pipelines laid with too much tension✓
 - Bends putting too much strain on the tubes✓
 - Joints not correctly fitted resulting in air leaks✓
- (3 × 1) (3)
- 5.3 Systematic maintenance helps to prolong service life✓ and improve the functional reliability of pneumatic control systems.✓
- (2 × 1) (2)

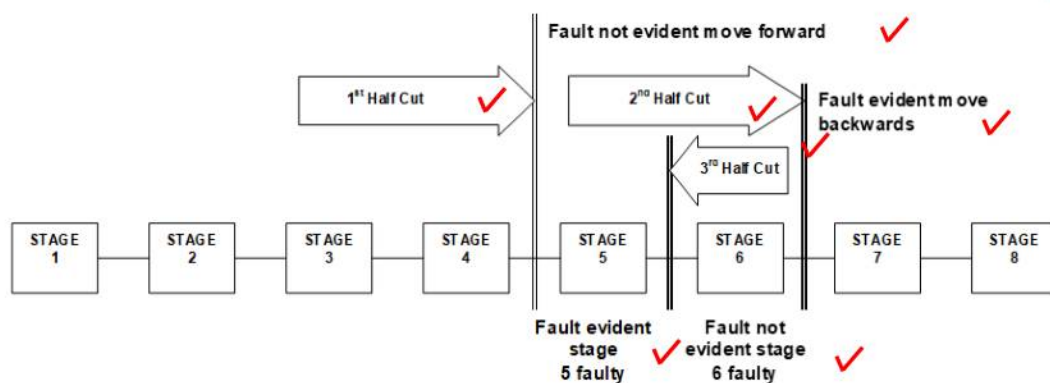
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MECHATRONIC SYSTEMS L2
(Second paper)

- | | | | | |
|-----|-------|---|---------|-----|
| 5.4 | 5.4.1 | Used to start, halt, interrupt or finish a sequence of flow✓ | (1 × 1) | (1) |
| | 5.4.2 | Used to indicate a process or operation✓ | (1 × 1) | (1) |
| | 5.4.3 | Used to indicate a decision that can only result in a yes or a no answer✓ | (1 × 1) | (1) |

5.5



(7 × 1) (7)

- | | | | |
|-----|--------------------------|---------|-----|
| 5.6 | Sequence function chart✓ | (1 × 1) | (1) |
|-----|--------------------------|---------|-----|

[20]

TOTAL: 100